

Take Home Exam
2110572 2024/2 NLP SYS
10% of your final grade
Kaggle end date: 16 March 11.00pm
Report due date 17 March 11.59pm
CU Automatic Short Answer Scoring



In this take home exam, your task is to create an automatic short answer scorer. This question is a competition format. The competition will be held until **16 March 2023**. The deadline for the submission will be at **11.00 pm, 16 March 2023**.

This is a private competition, you must click the following link to enter the submission page
[link](#)

You are required to submit a one-page report and code documentation at **midnight** 17 March.

- Your best model on Ipython notebook with outputs from all Ipython cells (**do not clear outputs**)

- A **1-sheet A4 report** which describe your model (**diagram included**), methodology and results. Include your kaggle name as appeared on the leaderboard in your report.

The maximum score for this competition is **20** points (10% of your grade). The scoring criteria is as follows:

1. Mean squared error (MSE) (10 points)

- MSE measures the average squared difference between predicted and actual values, with lower values indicating better performance.
- You can find more info about MSE here
<https://www.geeksforgeeks.org/mean-squared-error/>
- Your score for this section will be according to the MSE of the **private leaderboard** according to the formula:
 - $$\text{Score} = \text{MAX}(\text{MIN}(\frac{(\text{Weak_Baseline} - \text{Your_MSE}) \times 10}{(\text{Weak_Baseline} - \text{TA_Baseline})}, 10), 0)$$
- **The score on the private leaderboard differs from the public leaderboard.**
- **You can only submit 10 times per day**

1. Report (5 points)

- Your report should describe the details of your model and the methodology (how is your model working, what additional techniques do you use). Write clearly and concisely.
- Your report should also include a screenshot of your best submission on the leaderboard (one of the two that you selected).

2. Ranking (5 points)

- Score will be given by your rank in percentile (Extra points count as extra credits for the midterm exam). This is ranked within each cohort (undergrads, grads, etc)
 - Percentile [1 - 5] : 11 points
 - Percentile [6 - 10] : 10 points
 - Percentile [11 - 15] : 8 points
 - Percentile [16 - 20] : 6 points
 - Percentile [21 - 40] : 5 points
 - Percentile [41 - 60] : 3 points
 - Percentile [61 - 80] : 1 points
 - Percentile [81 - 100] : 0 points

We encourage you to use any models or pretrained weights or LLM API from the internet. However, **you are NOT allowed to use any additional data from other sources, or else you will not receive any score for this section of the midterm. You are not allowed to use the test set in any way besides for evaluation. For example, you cannot use the test set to learn students' answer patterns. You can use any off-the-shelf tools. Thus, you are encouraged to survey some techniques that can be tried to improve your performance.**

****You shall not share the data or model with anyone (inside or outside of class). This is an individual effort. You shall not discuss your solution with anyone.****

You must select two submissions as your main submissions. The best scoring one will be used to calculate your score.

The given files in kaggle competition are list as follows:

1. **train.csv** - The training data. Their columns are ID, set, question, answer, score. There are 4 questions in this dataset, which it was already provided in the question column. The answer column contains the students' answer with the score in the score column.
2. **test.csv** - The test data. There are 4 columns that are ID, set, question and answer. You have to submit the predictions according to the format of sample_submission.csv
3. **sample_submission.csv** - The sample of submission containing ID and score as columns. The predictions at the Predicted column are fixed for this file.

Your task is to predict the score of students' answers and submit them for evaluation. The submission requires 90 samples. The submitted file must be a .csv file.

The TA spent around 2 hours to get a MSE of 2.23550 on the public leaderboard. Plan accordingly!

Note that we do not limit the type of models used. Be sure to start simple and work your way up.